

CHAPTER 3: RESEARCH METHODOLOGY Guidelines (BSIT & BECE programmes)

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3.0 Introduction

This chapter describes the research methodology adopted in the study. It explains the research methods used to investigate the research problem, the rationale for selecting these methods, and how the research will be conducted. The chapter outlines the procedures used for data collection, system development, evaluation, and analysis. The methodology provides a structured framework that ensures the research objectives are systematically addressed and the results obtained are valid, reliable, and reproducible.

In computing research, different methodological approaches may be used depending on the nature of the problem. These approaches include empirical experiments, computer simulations, theoretical analysis, design science, artificial intelligence techniques, data mining, system design and implementation, action research, ethnography, process analysis, case studies, and econometric analysis.

Example:

The objective of this chapter is to explain **how the research problem identified in Chapter 1 will be addressed and how the proposed solution will be developed and evaluated.**

The chapter is organised as follows:

- Section 3.1 explains the research design.
- Section 3.2 describes the system development methodology.
- Section 3.3 presents the system architecture.
- Section 3.4 describes the data collection methods.
- Section 3.5 outlines the tools and technologies used.
- Section 3.6 explains the system implementation process.
- Section 3.7 describes the system testing and evaluation methods.
- Section 3.8 describes the ethical considerations.
- Section 3.9 describes the limitations of the methodology

3.1 Research Design

Research design refers to the **overall strategy used to integrate the different components of the study coherently and logically**. It ensures that the research problem is addressed effectively.

In computing research, the research design often combines several approaches, including:

- Experimental research
- Design science research
- Simulation-based research
- Case study analysis
- Data-driven research

The choice of research design depends on the nature of the research problem, the type of system being developed, and the objectives of the study.

Example:

The study adopts a **Design Science Research (DSR) approach**, which is commonly used in computing and information systems research.

Design Science Research focuses on **developing and evaluating technological artefacts** designed to solve identified problems. In computing research, these artefacts may include:

- software applications
- algorithms
- information systems
- network architectures
- decision support tools

In this study, the artefact developed is **[describe the system or application being developed]**.

The Design Science Research process typically involves the following stages:

1. **Problem Identification**
Defining the research problem and its significance.
2. **Definition of Objectives for the solution**
Identifying the requirements that the proposed system must satisfy.
3. **Design and Development of the artefact**
Designing and implementing the system architecture.

4. **Demonstration of the artefact**

Showing that the developed system can solve the problem.

5. **Evaluation of the artefact**

Assessing the performance and effectiveness of the system.

6. **Communication of results**

Presenting the research findings and contributions.

The Design Science approach is appropriate for this study because the research focuses on **developing and evaluating a computing system to solve a practical problem**

3.2 Research Methods

This section explains the research methods used in the study.

3.2.1 Empirical / Experimental Method

The empirical method is a research approach that concludes **observable and measurable evidence**. In computing research, empirical methods are commonly used to **evaluate systems, algorithms, and technologies through experimentation**.

Researchers conduct controlled experiments to test hypotheses and evaluate system performance.

This method is commonly applied in areas such as:

- Artificial neural networks (e.g., implementing new social networks using concepts from existing networks)
- Automated theorem proving
- Natural language processing
- Performance and behaviour analysis of systems
- Human–Computer Interaction (HCI)

Experimental evaluation is typically divided into **two phases**:

1. Exploratory Phase

During this phase, the researcher collects measurements and observations to identify important questions about the system under investigation.

2. Evaluation Phase

In this phase, experiments are conducted to answer the questions identified in the exploratory phase.

A well-designed experiment begins with a **clearly defined list of research questions** that the experiment aims to answer.

Experiments may be conducted to study the behaviour of:

- Natural systems (e.g., physics-based simulations)
- Humans and organizations (e.g., experimental psychology)
- Artificial systems (e.g., algorithm performance evaluation)

Experiments may involve:

- Performance benchmarking
- Usability testing
- System comparison

3.2.2 Computer Simulation Method

The **computer simulation method** is used to study complex phenomena that may be difficult, impractical, or too expensive to investigate through real-world experimentation.

Computer simulations allow researchers to **model systems and evaluate their behaviour under different conditions**.

This method is particularly useful for investigating systems that are outside the experimental domain.

Common application areas include:

- Computer networks (e.g., developing and testing new communication protocols)
- Computer hardware architectures
- Traffic flow simulation
- Distributed systems modelling
- Virtual reality simulations

Through simulation models, researchers can evaluate system performance, predict behaviour, and analyse outcomes without implementing the system in a real-world environment.

Simulation tools such as **NS-3, OMNeT++, MATLAB, or Python simulation frameworks** are often used.

3.2.3 Theoretical Method / Formal Methods and Mathematical Proof

The theoretical research method is based on **logic, mathematics, and formal reasoning**. It is used to develop **algorithms, computational models, and formal proofs**.

In this approach, researchers use mathematical techniques to demonstrate that a system possesses certain desired properties.

Typical outputs of theoretical research include:

- Conceptual models
- Mathematical models
- Algorithms
- Formal proofs

Common techniques used include:

- Iteration
- Recursion
- Complexity analysis
- Mathematical induction

For example, when developing a new **machine learning classifier**, researchers may use mathematical modelling to prove the efficiency and correctness of the model.

Theoretical methods are widely used in areas such as:

- Computational complexity theory
- Algorithm design
- Formal logic systems
- Artificial intelligence theory

3.2.4 Design Science

Design science is a research methodology commonly used in **information systems and software engineering**. The primary objective is to **create innovative artefacts that solve practical problems**.

Artefacts produced in design science research may include:

- Software systems
- Algorithms
- Data models

- Hardware models
- Databases
- Frameworks
- Methodologies

Design science research is often referred to as **constructive research** because it involves building and evaluating new solutions based on theoretical principles.

A **build methodology** involves constructing an artefact to demonstrate that a solution is feasible.

For the research to qualify as a contribution, the artefact must:

- Introduce an innovation, or
- Contain **new features not previously demonstrated**

Design science emphasises **systematic evaluation and iterative improvement** of artefacts to ensure they meet intended objectives.

The design science process usually involves:

1. Problem identification
2. Artefact design
3. Artefact development
4. Artefact evaluation
5. Iterative improvement

This methodology has become increasingly important as organisations seek improved information systems design to avoid costly IT failures and poor data management.

3.2.5 Artificial Intelligence Method

Artificial Intelligence (AI) research methods involve the use of **intelligent algorithms and computational models** to solve complex problems.

Common AI techniques include:

- Machine learning
- Neural networks
- Expert systems
- Intelligent agents
- Natural language processing

AI techniques can automate tasks such as:

- Data processing
- Predictive analytics
- Pattern recognition
- Automated Decision support systems

Agent technologies can also be used to automate processes such as:

- Source selection
- Data conversion
- Predictive search
- Intelligent user interaction
- Intelligent recommendation systems

AI methods are particularly useful in improving **system performance, decision support systems, and user experience.**

3.2.6 Data Mining

Data mining is the process of **extracting hidden, previously unknown, and potentially useful information from large datasets.**

It integrates methods from:

- Database systems
- Machine learning
- Artificial intelligence
- Statistics

Typical data mining tasks include:

- Classification
- Clustering
- Association rule mining
- Anomaly detection

Data mining is widely applied in:

- Business analytics
- Fraud detection
- Customer behaviour analysis

- Cybersecurity

Data mining techniques can also be applied to address various data quality issues.

For example:

- **Outlier detection algorithms** can identify anomalies in datasets.
- **Data cleansing techniques** can improve data quality.
- **Intrusion detection systems** can identify suspicious activities.

Data mining is also widely used in **schema matching**, where algorithms identify relationships between different database schemas.

Although many data mining algorithms are robust, special handling may be required when working with datasets containing data quality issues.

3.2.7 System Design and Implementation

The system design and implementation method is widely used in **software engineering and information systems research**.

This method involves:

- System analysis
- System design
- System implementation
- System testing

Researchers use this approach to develop **proof-of-concept systems** that demonstrate the feasibility of proposed solutions.

This methodology typically draws upon principles from:

- Software engineering
- Database design
- Data modelling
- System architecture

The developed system is then evaluated to determine whether it meets the research objectives.

The implementation tools may include:

- Programming languages (Python, Java, C++)
- Databases (MySQL, PostgreSQL)
- Development frameworks
- Cloud platforms

3.3 System Architecture

This section describes the **overall structure of the proposed system**.

It includes:

- System components
- Data flow
- User interfaces
- Hardware and software components

A system architecture diagram should be included to illustrate the interaction between system modules.

Example:

The proposed system follows a **[insert architecture type]** architecture.

Examples include:

- Client–Server Architecture
- Three-Tier Architecture
- Cloud-Based Architecture
- IoT-Based Architecture

The architecture consists of the following components:

1. User Interface Layer

Provides interaction between users and the system through graphical interfaces.

2. Application Logic Layer

Handles system processing, algorithms, and business logic.

3. Data Storage Layer

Stores system data using a database management system.

A conceptual diagram of the system architecture is presented in **Figure 3.1**.

(Students must insert a system architecture diagram here.)

3.4 Data Collection Methods

This section describes how data will be collected for the study. Data collection is necessary to support the development and evaluation of the proposed system.

Possible data sources include:

- System logs
- User interaction data
- Surveys or questionnaires
- Existing datasets
- Sensor data
- Public data repositories

Data collection methods must ensure **data reliability and validity**.

Example:

The following data sources are used in this study.

3.4.1 Literature Review

Relevant academic literature was reviewed to identify existing technologies, algorithms, and systems related to the research problem.

Sources included:

- academic journals
- conference papers
- technical reports
- scholarly databases

3.4.2 Dataset Collection

The system uses **[describe dataset source if applicable]**, which may include:

- publicly available datasets
- system-generated data
- simulated data
- user input data
- sensor data (for IoT systems)

3.4.3 System Requirements Gathering

User requirements were obtained through:

- interviews with potential users
- observation of existing systems
- analysis of system documentation

3.5 Tools and Technologies Used

The development of the proposed system utilised several software and hardware tools.

3.5.1 Programming Languages

Examples may include:

- Python
- Java
- C++
- JavaScript

3.5.2 Development Frameworks

Examples include:

- TensorFlow / PyTorch (AI systems)
- Node.js / Django or Flask(web systems)
- React or Angular (frontend development)
- Flutter / React Native (mobile applications)

3.5.3 Database Management Systems

The system uses [MySQL / PostgreSQL / MongoDB / SQLite] for data storage.

3.5.4 Development Environment

The system was developed using:

- Visual Studio Code
- PyCharm
- Eclipse
- Android Studio

3.6 System Implementation

The implementation phase involved translating the system design into a functional application.

Key activities included:

- coding system modules
- integrating system components
- database implementation
- user interface development

The system was implemented following the architecture described in Section 3.4.

3.7 System Testing & Evaluation

System testing is conducted to ensure that the system meets its functional and performance requirements.

3.7.1 Types of Testing Conducted

Unit Testing

Individual modules were tested independently to verify their functionality.

Integration Testing

System modules were combined and tested to ensure they interact correctly.

System Testing

The complete system was tested to verify that it satisfies all functional requirements.

3.7.2 Evaluation Methods

Evaluation determines whether the proposed system meets the research objectives.

Evaluation techniques may include:

Performance Evaluation

Measures system performance using metrics such as:

- Execution time
- Throughput
- Memory usage
- system accuracy
- response time
- processing speed

- system reliability
- Network latency

Where applicable, the proposed system is compared with existing systems to determine performance improvements.

Accuracy Evaluation

Used in AI and machine learning systems.

Common metrics include:

- Accuracy
- Precision
- Recall
- F1-score

Usability Evaluation

Used to evaluate user experience.

Methods include:

- User testing
- Surveys
- Task completion analysis

Comparative Analysis

Compares the proposed system with existing systems.

3.8 Ethical Considerations

This section explains how ethical issues are addressed during the research.

Key considerations include:

- User Data privacy
- Informed consent
- Responsible data usage
- Avoidance of bias in algorithms
- Protection of participant information

Ethical approval may be required when research involves **human participants or personal data**.

3.9 Limitations of the Methodology

This section discusses potential limitations of the research methodology.

Examples include:

- Limited dataset size
- Simulation constraints
- Hardware limitations
- Restricted access to real-world data

Acknowledging limitations helps provide transparency in the research process.

3.10 Chapter Summary

This chapter presented the research methodology used in the study. It described the research design, methods used for system development, data collection procedures, tools used in implementation, and the evaluation methods used to assess the proposed system.

Example:

This chapter described the research methodology used in the development of the proposed system. The study adopted a **Design Science Research approach**, supported by the **System Development Life Cycle methodology**.

The chapter also described the system architecture, data collection methods, development tools, and evaluation techniques used in the research.

The next chapter presents the **system design**.

A good Chapter 3 must answer five questions clearly:

1. **What research approach did you use?**
(Design science, experiment, simulation)
2. **How will the system be developed?**
(SDLC, Agile, prototype model)
3. **What tools will you use?**
4. **What data will the system use?**
5. **How will you evaluate the system?**